

Direct Numerical Simulations of Mixed-Phase Cloud Glaciation Caused by Turbulent Mixing

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Key Points:

- More but smaller ice crystals promote glaciation compared to fewer but larger ice crystals for the same initial ice mass.
- Turbulence enhances glaciation if the ice number density is high but has no effect on glaciation if the ice number density is low.
- Hexagonal ice crystals grow preferentially along the shorter dimension and their non-isotropic growth suppresses glaciation compared to spherical ice crystals.
- The size distributions of both cloud droplets and ice crystals are skewed to the left during glaciation if the turbulence intensity is low.

Abstract

Glaciation of turbulent mixed-phase clouds is examined at the microscale using direct numerical simulations and Lagrangian particle tracking. The effects of flow turbulence intensity, ice number density, and ice crystal aspect ratio on glaciation are investigated. The results show that turbulence enhances glaciation if the ice number density is high. When the ice number density is low, variation of turbulence intensity does not appear to affect the rate of glaciation. For the same initial ice mass, smaller but more ice crystals promote glaciation compared to larger but fewer ice crystals. It is found that hexagonal ice crystals grow preferentially along the shorter dimension and their non-isotropic growth suppresses glaciation relative to spherical ice crystals. Turbulent mixing affects both the mean and fluctuating thermodynamic fields in our model and glaciation depends on the microscale details of turbulence-cloud-ice crystal interactions.

1 Introduction

The saturation vapor pressure with respect to liquid water is higher than that with respect to ice. If the environmental supersaturation level is such that it is supersaturated with respect to ice but subsaturated with respect to water, ice crystals will grow while cloud droplets will shrink. If this continues, a mixed-phase cloud will turn into an ice cloud or glaciates (Korolev et al., 2003). This is the Wegener-Bergeron-Findeisen (WBF) process (Storelvmo & Tan, 2015), also commonly known as glaciation. But mixed-phase clouds have been found to be stable (Morrison et al., 2012), meaning that they do not fully glaciates and remain in a mixed phase condition. To understand the glaciation and persistence of mixed-phase clouds, the interactions between ambient turbulence, cloud droplets, and ice crystals need to be studied at the microscale.

The WBF mechanism was studied analytically by Korolev (2007) who showed that water vapor mass can increase or decrease during this process. An increase of water vapor means that evaporation of cloud droplets is faster than deposition of ice crystals while a decrease of water vapor indicates the opposite. The analytical investigation by Pinksly et al. (2014) suggested that ice crystal number density and initial liquid water mass strongly influence the rate of glaciation. Mixed-phase clouds had been studied numerically. Hill et al. (2013) studied the growth of cloud and ice using a large eddy simulation (LES) model. The authors reported that liquid water mass in a mixed-phase cloud decreases if the ice number density and ice crystal radius increase. Yang et al. (2015) applied LES to examine the growth of ice crystals in a mixed-phase cloud and found that a fraction of ice crystals get trapped in large eddies and grow much bigger than the rest. In their LES investigation, Chen et al. (2024) studied the evolution of mixed-phase clouds using direct numerical simulations (DNS) to report that the size distributions of cloud droplets broaden more compared to those of ice crystals. They further suggested that initial liquid mass should be large enough for mixed-phase condition to persist. Sarnitsky et al. (2024) used DNS to investigate mixed-phase cloud and reported that flow turbulence affects ice crystal growth and thus glaciation.

The geometry of ice crystals is important, particularly at the microscale. Using a theoretical approach, Chen and Lamb (1994) showed that aspect ratio (height/base) of ice crystals changes as ice mass grows with time. In a laboratory investigation, Bailey and Hallett (2003) showed that the complexity of shape for plate-like ice crystals increases if the supersaturation with respect to ice is higher. Sheridan et al. (2009) suggested that growth is more isometric if ice crystals are larger initially but more twisted if they are smaller. Sulia and Harrington (2011) found that glaciation rate can be under-predicted by nearly an order of magnitude if equivalent spheres are assumed in place of precise ice crystal shape.

In the present study, we implement DNS with Lagrangian tracking of cloud droplets and ice crystals to study glaciation of mixed-phase clouds. For this purpose, we modify our previous DNS model (Sharfuddin et al., 2025) by including the ice phase equations. The cloudy and clear-air regions are initially separated by a sharp interface and subsequent turbulent mixing is simulated. We vary the intensity of flow turbulence and number density of ice crystals to assess their impacts on glaciation. The growth profiles of spherical and hexagonal ice crystals are also compared.

2 Governing Equations

The turbulent flow of air is governed by the incompressible Navier-Stokes equations:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here, $\mathbf{u}$ is the instantaneous velocity vector, ν is the kinematic viscosity, p is the instantaneous pressure and ρ_a is the density of air. $\mathbf{F}_c$ is an external force to sustain turbulence as described in Sharfuddin et al. (2025). The transport equations for the thermodynamic fields of temperature (T) and water vapor mixing ratio (q_v) are:

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \frac{L_w}{c_p} C_w + \frac{L_{ice}}{c_p} C_{ice} + \mu_T \nabla^2 T, \quad (3)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla) q_v = -(C_w + C_{ice}) + \mu_v \nabla^2 q_v. \quad (4)$$

Above, L_w and L_{ice} are specific latent heats of condensation and deposition, respectively. C_w is the time rate of mass exchange between vapor and liquid while C_{ice} denotes the same between vapor and ice. Also, μ_T and μ_v are molecular diffusivities of temperature and vapor, respectively. The environmental supersaturation (S_e) depends on the saturation vapor mixing ratio ($q_{v,s}$) and the saturation vapor pressure (p_{sat}).

$$S_e = \frac{q_v}{q_{v,s}} - 1 = \frac{q_v}{621.97 \frac{p_{sat}}{p - p_{sat}}} - 1, \quad (5)$$

p_{sat} can be calculated with respect to water and ice (Huang, 2018).

$$p_{sat} = \begin{cases} p_{sat,w} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right) \\ p_{sat,ice} = 611.2 \times \exp\left(\frac{22.46(T - 273.15)}{T - 0.53}\right) \end{cases}. \quad (6)$$

The cloud droplets and ice crystals are tracked individually in Lagrangian fashion. The equations for the position ($\mathbf{X}$) and velocity ($\mathbf{V}$) vectors can be written as:

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad \frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}, \quad (\tau_p)_i = \frac{2\rho_i(r_i)^2}{9\rho_a\nu}. \quad (7)$$

where the subscript ' i ' denotes the i -th particle (cloud droplet or ice crystal). $\mathbf{u}_i$ is the instantaneous fluid velocity vector interpolated at the position of particle i and τ_p is the particle response time.

Conveniently, the S_e and particle radius (r) can be written as:

$$S_e = \begin{cases} S_{e,w} & \text{if } p_{sat} = p_{sat,w} \\ S_{e,ice} & \text{if } p_{sat} = p_{sat,ice} \end{cases}, \quad r = \begin{cases} r_w & \text{if } S_e = S_{e,w} \\ r_{ice} & \text{if } S_e = S_{e,ice} \end{cases}. \quad (8)$$

The growth equations for cloud droplets and ice crystals can be written as:

$$\frac{d(r_w(t))_i}{dt} = \left(\frac{G_w}{r_w(t)} S_{e,w}(\mathbf{X}, t) \right)_i, \quad \frac{d(r_{ice}(t))_i}{dt} = \left(\frac{G_{ice}}{r_{ice}(t)} S_{e,ice}(\mathbf{X}, t) \right)_i. \quad (9)$$

88 G_w and G_{ice} are expressed as:

$$G_w = \frac{1}{\frac{L_w \rho_w}{k_T T} \left(\frac{L_w}{R_v T} - 1 \right) + \frac{\rho_w R_v T}{\mu_v p_{sat,w}}}, \quad G_{ice} = \frac{1}{\frac{L_{ice} \rho_{ice}}{k_T T} \left(\frac{L_{ice}}{R_v T} - 1 \right) + \frac{\rho_{ice} R_v T}{\mu_v p_{sat,ice}}}. \quad (10)$$

89 where k_T is the thermal conductivity in air and R_v is the specific gas constant for water vapor.
90 With the foregoing, C_w and C_{ice} can be determined as:

$$C_w(\mathbf{x}, t) = \frac{4\pi\rho_w}{\rho_a a^3} \sum_{i=1}^{n_w} \left(G_w r_w(t) S_{e,w}(\mathbf{X}, t) \right)_i, \quad (11)$$

$$C_{ice}(\mathbf{x}, t) = \frac{4\pi\rho_{ice}}{\rho_a a^3} \sum_{i=1}^{n_{ice}} \left(G_{ice} r_{ice}(t) S_{e,ice}(\mathbf{X}, t) \right)_i. \quad (12)$$

91 where ρ_w and ρ_{ice} are liquid and ice densities, respectively. n_w is the number of cloud droplets in
92 a grid cell with a volume of a^3 and n_{ice} is the same for ice crystals. Note that r_{ice} is the equivalent
93 radius of ice crystals if they are not spherical. Let us consider a regular hexagon shape with aspect
94 ratio of h/b , where h is the height and b is the base. Then r_{ice} is found as

$$\frac{4}{3}\pi r_{ice}^3 = \frac{3\sqrt{3}}{2} b^2 h, \quad r_{ice}^3 = 0.62(b^2 h), \quad r_{ice} = 0.8527(b^2 h)^{1/3}. \quad (13)$$

95 The ice crystal shape is hexagonal column if $h/b > 1$ and hexagonal plate if $h/b < 1$. We see in
96 Equation (9) that dr_{ice}/dt is inversely proportional to r_{ice} . Hence, the base will change at twice
97 the rate of height when $h/b = 2$. To account for different growth rates along the base and height
98 of hexagonal ice crystals, we solve:

$$\frac{db_i(t)}{dt} = \left(\frac{G_{ice}}{b(t)} S_{e,ice}(\mathbf{X}, t) \right)_i, \quad \frac{dh_i(t)}{dt} = \left(\frac{G_{ice}}{h(t)} S_{e,ice}(\mathbf{X}, t) \right)_i. \quad (14)$$

99 Equation (14) gives the updated b and h . Then r_{ice} is calculated using Equation (13). It is assumed
100 that any changes in the $S_{e,ice}$ are negligible at the μm scale. So, $S_{e,ice}$ is calculated at the center of
101 each ice crystal. Moreover, ice crystals grow solely through diffusion in our model and processes
102 like ice nucleation, secondary ice production, and ventilation are neglected.

103 3 Numerical Details

104 To vary the flow turbulence level, we force different wavenumber bands as discussed in
105 Sharfuddin et al. (2025). Specifically, we use the wavenumber bands: $1 \leq k_f/k_{min} \leq 3.0$ and $1 \leq$
106 $k_f/k_{min} \leq 16.18$ to generate ‘low’ and ‘high’ turbulence levels and denote them as ‘L’ and ‘H’,
107 respectively. k_f is the forced wavenumber and k_{min} is the minimum wavenumber. The dissipation
108 rates of turbulent kinetic energy (TKE) are $0.01388 \text{ m}^2/\text{s}^3$ and $0.00037 \text{ m}^2/\text{s}^3$ and the Kolmogorov
109 length scales are 0.00092 m and 0.00226 m for ‘low’ and ‘high’ turbulence levels, respectively.

110 The physical model simulated is that of a cubic domain inside which turbulence, cloud
111 droplets, and ice crystals interact. The domain volume is 0.512^3 m^3 and the boundary conditions
112 are triply periodic. At the start of simulations, the domain is divided equally into cloudy and clear-
113 air regions. Cloud droplets and ice crystals are placed in the cloudy region only and subsequent
114 turbulent mixing causes them to spread to all over the domain. A slab-like setup is assumed for the

initial field to enable a sharp transition in the vapor mixing ratio and temperature at the cloud/clear-air interface (Sharfuddin et al., 2025). The initial vapor mixing ratio field is:

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & \text{elsewhere} \end{cases} \quad (15)$$

Above, $q_v^{max} = 2.367 \text{ g/kg}$, $q_{v,e} = 1.90 \text{ g/kg}$, and $d \equiv L/2$ is the width of the cloud slab. The temperature field is initialized as:

$$T(t = 0) = \langle T(t = 0) \rangle - 0.608 \langle T(t = 0) \rangle (q_v(z, t = 0) - \langle q_v(t = 0) \rangle). \quad (16)$$

We simulate six cases: L-H, H-H, L-L, H-L, L-H-C, and L-H-P, as shown in Table I. The table also lists the values of relevant parameters. Note that higher ice number density corresponds to smaller mean ice crystal radius and the vice versa in our model because initial ice masses are equal for all cases. Ice crystals are spherical for the first four cases, hexagonal column for Case L-H-C and hexagonal plate for Case L-H-P. The cloud droplet number density is 59.6 cm^{-3} for all cases and is much higher than that of ice crystals which is realistic for mixed-phase clouds. The number of grid points is 256^3 for all cases. Particles (cloud droplets or ice crystals) are inserted in the model domain after 4 s when turbulence becomes statistically stationary.

The present setup differs from that of Chen et al. (2024) since our model domain is initially divided into two regions with distinct supersaturation levels. By allowing the regions to mix, the effects of turbulent mixing are evaluated. Also, our DNS model uses a fifth order finite difference scheme to solve the convective terms in Equations (1), (3), and (4) as sharp interfaces are present (Equations (15) and (16)). Chen et al. (2024) applied a pseudospectral scheme.

Table 1

Details of the test cases

Case	TKE dissipation rate (m^2/s^3)	Ice number intensity (cm^{-3})	Initial ice crystal radius (μm)	Initial cloud droplet radius (μm)	Aspect ratio (h/b) of ice crystals
L-H	0.01388	0.011	7.5	7.5	1.0
H-H	0.00037	0.011	7.5	7.5	1.0
L-L	0.01388	0.0011	16.16	7.5	1.0
H-L	0.00037	0.0011	16.16	7.5	1.0
L-H-C	0.01388	0.011	7.5	7.5	2.5
L-H-P	0.01388	0.011	7.5	7.5	0.4
Case	Comments				
L-H	low (L) turbulence level, high (H) ice number density, spherical				
H-H	high (H) turbulence level, high (H) ice number density, spherical				
L-L	low (L) turbulence level, low (L) ice number density, spherical				
H-L	high (H) turbulence level, low (L) ice number density, spherical				
L-H-C	low (L) turbulence level, high (H) ice number density, hexagonal column (C)				
L-H-P	low (L) turbulence level, high (H) ice number density, hexagonal plate (P)				

<i>Parameters and their values</i>					
Symbol	Value	Unit	Symbol	Value	Unit
L_w	2.50×10^6	J/kg	c_p	1005.0	J/kg/K
L_{ice}	2.83×10^6	J/kg	ρ_a	0.7674	kg/m ³
ρ_w	1000	kg/m ³	ρ_{ice}	917	kg/m ³
p	57160	N/m ²	$\langle T(t = 0) \rangle$	259.53	K
R_v	461.5	J/kg/K	k_T	0.0238	W/m/K
μ	1.64×10^{-5}	m ² /s	$\mu_T = \mu_v$	2.2×10^{-5}	m ² /s

4 Results and Discussion

The mean supersaturations with respect to water and ice at initial time are negative and positive, respectively. Hence, ice crystals grow but cloud droplets evaporate, and glaciation takes place. We vary the flow turbulence level, ice number density, and ice crystal aspect ratio. Their effects on thermodynamics and microphysics are discussed.

Figures 1(a)-(d) show the temporal evolutions of mean thermodynamic fields. Liquid water converts into water vapor during evaporation of cloud droplets and ice crystals grow by absorbing water vapor. So, water vapor increases and decreases at the same time but there is a net increase of water vapor (Figure 1(b)). This is because evaporation is dominant over ice deposition as cloud droplet mass exceeds that of ice crystals. The increase of $\langle q_v \rangle$ is accompanied by a decrease of $\langle T \rangle$ because more energy is absorbed through evaporation than released through deposition. The $\langle S_{e,w} \rangle$ and $\langle S_{e,ice} \rangle$ are directly proportional to the $\langle q_v \rangle$ and increase similarly (Figures 1(c)-(d)). The time evolutions of mean profiles appear to be steady when the $\langle S_{e,w} \rangle$ approaches zero (saturation). The increase of $\langle S_{e,w} \rangle$ from negative to zero is fastest for Case H-L but slowest for Case H-H (Figure 1(c)). The longer the subsaturation (negative $S_{e,w}$) persists, the stronger is the evaporation. Thus, more ice but smaller crystals enhance evaporation in our model if the turbulence level is high (Case H-H). However, the $\langle S_{e,w} \rangle$ profiles are identical for Cases L-H and L-L, suggesting that ice number density (or ice crystal size) does not influence evaporation if the turbulence level is low. Between Cases H-L and H-H, the decrease of $\langle T \rangle$ and increase of $\langle q_v \rangle$ are in the opposite order. This implies that less heat is absorbed during combined evaporation and deposition for Case H-L while more water vapor is produced for this case. Had we simulated warm clouds, the increase of $\langle q_v \rangle$ would be proportional to the decrease of $\langle T \rangle$ because the magnitudes of the latent heat of condensation and evaporation per unit mass are equal. Whereas ice phase is present in mixed-phase clouds, and the latent heat of deposition is higher than that of evaporation (Table 1).

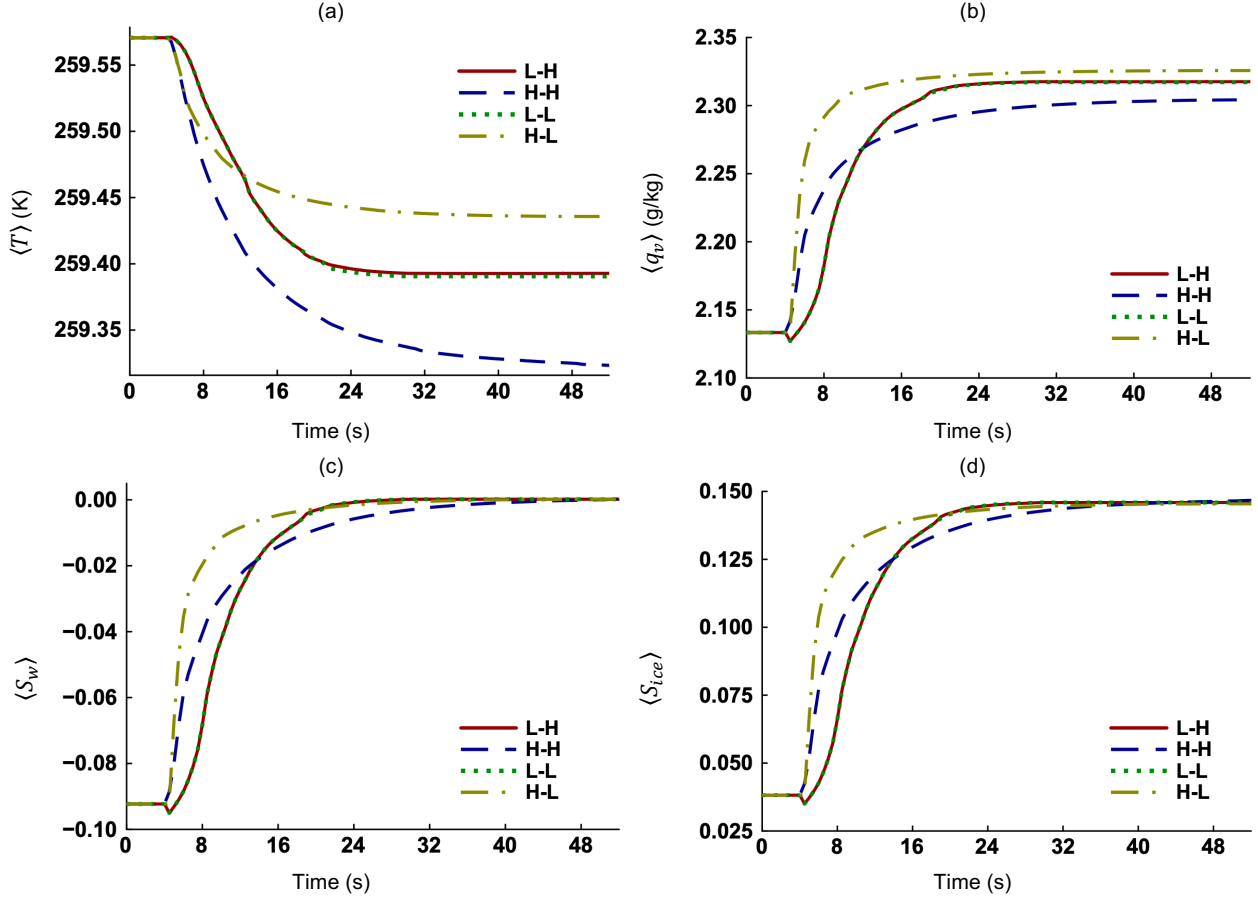
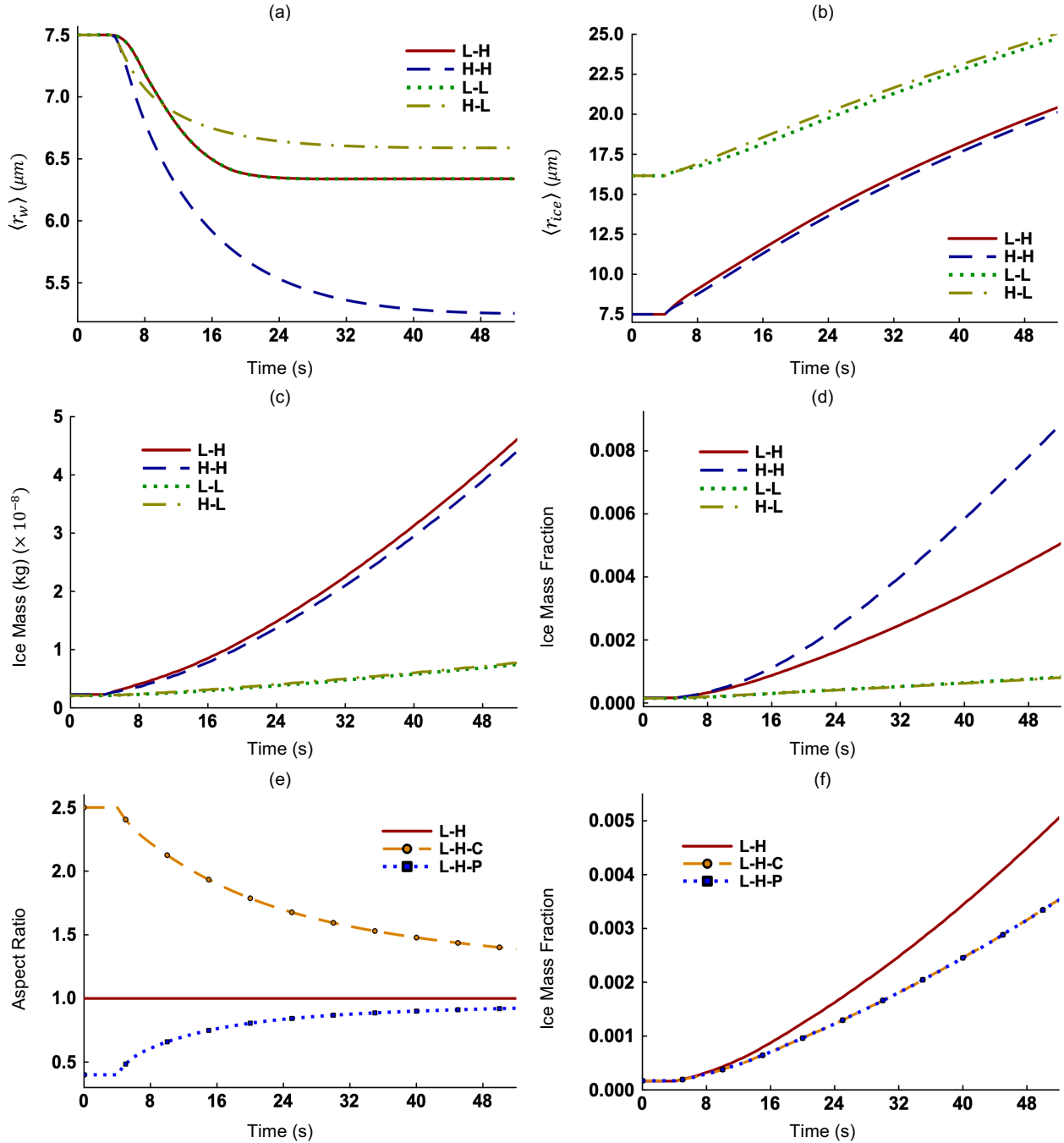


Figure 1. Time evolution of the (a) mean temperature ($\langle T \rangle$), (b) mean water vapor mixing ratio ($\langle q_v \rangle$), (c) mean environmental supersaturation with respect to water ($\langle S_{e,w} \rangle$) and (d) mean environmental supersaturation with respect to ice ($\langle S_{e,ice} \rangle$).

Consistent with Figure 1(c), stronger evaporation leads to a maximum decrease of $\langle r_w \rangle$ for Case H-H (Figure 2(a)) and weaker evaporation corresponds to a minimum decrease of $\langle r_w \rangle$ for Case H-L. The $\langle r_{ice} \rangle$ increases almost linearly for the four cases (Figure 2(b)) as the environment is highly supersaturated with respect to ice (Figure 1(d)). Also, the $\langle r_{ice} \rangle$ increases slightly faster for Case L-H compared to Case H-H, indicating that high turbulence has a retarding effect on ice growth if ice crystals are smaller but more in number. The converse is true if there are fewer but larger ice crystals (Cases L-L vs H-L). The masses of ice crystals are initially equal for the four cases as shown in Figure 2(c) and they keep increasing as expected during glaciation. The increase is more significant when ice number density is lower and ice crystals are larger (Cases L-H and H-H) compared to the opposite (Cases L-L and H-L). It is evident that number density and mean radius of ice crystals play a more important role during ice growth than flow turbulence does. The ice mass fraction, which is the ratio of ice mass and total mass of liquid and ice, increases most for Case H-H (Figure 2(d)). Although ice growth is slightly higher for Case L-H compared to Case H-H (Figure 2(c)), significant evaporation for the latter results in a higher ice mass fraction. Overall, fewer but larger ice crystals slow down glaciation while more but smaller ice crystals promote

175 glaciation (Figure 2(d)). The same plot also shows that flow turbulence enhances glaciation if the
 176 ice number density is high.



177 **Figure 2.** Time evolution of the (a) mean radius of cloud droplets ($\langle r_w \rangle$), (b) mean radius of ice
 178 crystals ($\langle r_{ice} \rangle$), (c) ice mass and (d) ice mass fraction, for Cases L-H, H-H, L-L, and H-L. Time
 179 evolution of the (e) ice crystal aspect ratio and (f) ice mass fraction, for Cases L-H, L-H-C, and
 180 L-H-P.

181 Figures 2(e)-(f) show the time evolutions of ice crystal aspect ratio and ice mass fraction,
 182 respectively, for three cases: L-H, L-H-C, and L-H-P. The initial aspect ratios are 1.0, 2.5, and 0.4
 183 for ice crystals resembling sphere, hexagonal column, and hexagonal plate, respectively. We find

that the aspect ratios of hexagonal column and plate converge to unity and that of sphere remains unchanged with time. Hence, ice crystals grow preferentially along the shorter dimension. As ice number density is much lower than that of cloud droplets in our model, ice crystals do not face competition for water vapor. With diffusional growth and availability of water vapor, ice crystals grow faster in the direction they are smaller. In mixed-phase clouds, ice crystals of complex shape exist depending on the temperature, supersaturation, and growth mechanism (Bailey and Hallett, 2003). But in absence of collisional breakup and branching, hexagonal ice crystals evolve to be spherical in our model. Figure 2(f) shows that ice mass fraction increases faster for Case L-H than for Cases L-H-C and L-H-P. Diffusional growth is more efficient when it is isotropic (spherical) instead of anisotropic (hexagonal plates and columns).

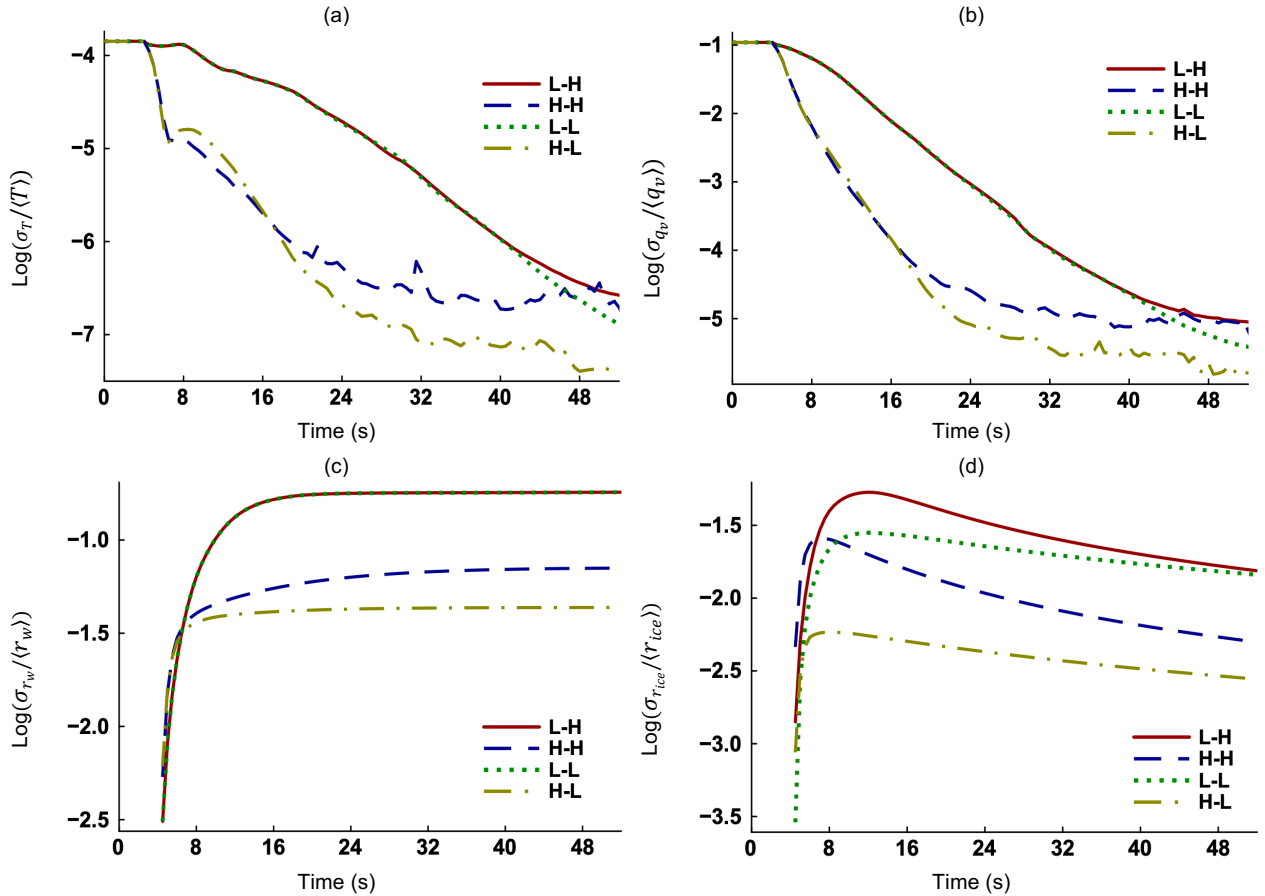


Figure 3. Time evolution of the relative dispersions of (a) temperature ($\sigma_T/\langle T \rangle$), (b) vapor mixing ratio ($\sigma_{q_v}/\langle q_v \rangle$), (c) cloud droplet radii ($\sigma_{r_w}/\langle r_w \rangle$), (d) ice crystal radii ($\sigma_{r_{ice}}/\langle r_{ice} \rangle$).

The time evolutions of the relative dispersion, which is standard deviation divided by mean, are plotted in Figures 3(a)-(b) for temperature and vapor mixing ratio, respectively. The magnitude of thermodynamic fluctuations is largest at the start when the T and q_v have distinct values in the two regions. Both $\sigma_T/\langle T \rangle$ and $\sigma_{q_v}/\langle q_v \rangle$ decrease afterwards since turbulent mixing homogenizes the thermodynamic fields (Sharfuddin et al., 2025). The higher the turbulence level, the quicker the homogenization and the reduction of fluctuations (Cases H-L and H-H). Figures 3(c)-(d) are analogous to Figures 3(a)-(b) but for the radii of cloud droplets and ice crystals, respectively. At early times, monodisperse cloud droplets and ice crystals are exposed to turbulent fluctuations and their relative dispersions increase. Later, the $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ decreases from its peak (Figure 3(d)) but

the $\sigma_{r_w}/\langle r_w \rangle$ becomes steady (Figure 3(c)) for all four cases. Although cloud droplet evaporation
 steadies (Figure 2(a)), ice crystals continue to grow (Figure 2(b)) and their relative dispersions
 decrease, consistent with declining turbulent fluctuations. We also observe that higher turbulence
 intensity has a retarding effect on the relative dispersions of both cloud droplets and ice crystals as
 a result of enhanced mixing. The ice number density has an expediting effect on the $\sigma_{r_{ice}}/\langle r_{ice} \rangle$
 for both turbulence levels and on the $\sigma_{r_w}/\langle r_w \rangle$ if the turbulence level is high (Cases H-H vs H-L).
 However, it has no effect on the $\sigma_{r_w}/\langle r_w \rangle$ when the turbulence level is low (Cases L-H vs L-L).

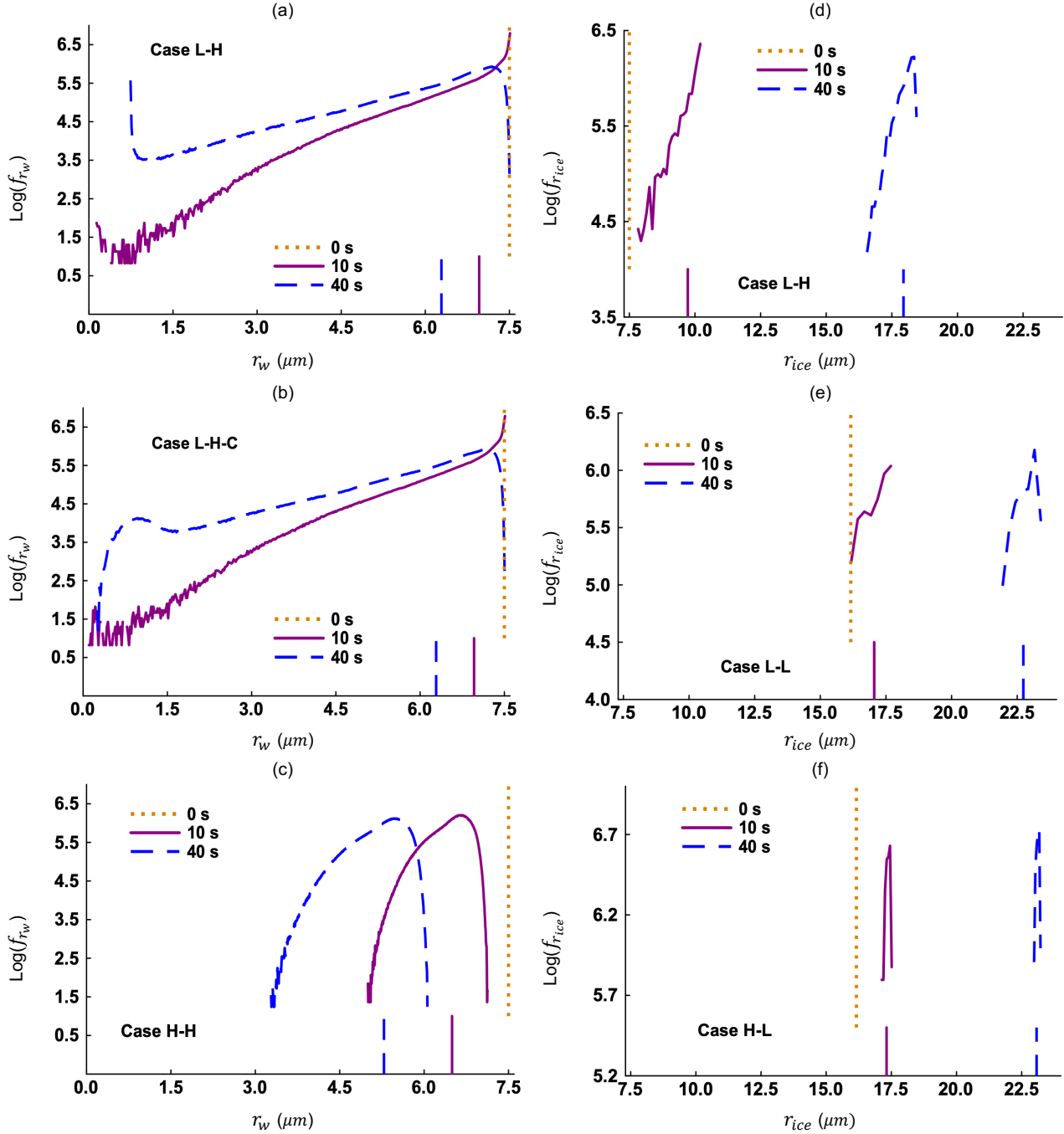


Figure 4. Log PDFs of cloud droplet size for Cases (a) L-H, (b) L-H-C, and (c) H-H. Log PDFs of ice crystal size for Cases (d) L-H, (e) L-L, and (f) H-L. Each plot includes PDFs at three times: 0 s, 10 s, and 40 s. The vertical line segments at the bottom indicate mean values in the respective distributions.

We examine the size distributions of particles in Figure 4. The PDFs of cloud droplet radii (f_{r_w}) are shown in Figures 4(a)-(c) and those of ice crystal radii ($f_{r_{ice}}$) are shown in Figures 4(d)-(f). Both cloud droplets and ice crystals follow monodisperse size ($7.5 \mu m$) distributions at initial time. The radius PDFs broaden by 10 s (Figures 4(a)-(f)) due to turbulent fluctuations. The f_{r_w} are strongly skewed to the left for Cases L-H and L-H-C at 10 s and 40 s. Hence, most cloud droplets experience partial evaporation and are relatively larger for both cases. A sudden spike in the left tail of f_{r_w} (Figure 4(a)) suggests the prevalence of very small cloud droplets for Case L-H at 40 s. Figure 4(b) shows that the f_{r_w} is bimodal for Case L-H-C at 40 s with one peak for very small cloud droplets and another for larger droplets. As higher flow turbulence corresponds to reduced microphysical fluctuations, the f_{r_w} are much narrower and less skewed to the left for Case H-H (Figure 4(c)) relative to Case L-H. The $f_{r_{ice}}$ are skewed to the left for Cases L-H and L-L (Figures 4(d)-(e)), indicating that larger ice crystals are prevalent. The leftward decrease of $f_{r_{ice}}$ from the peak is not strict when ice crystals are larger but fewer (Case L-L). Again, higher flow turbulence makes the $f_{r_{ice}}$ narrower (Figure 4(f)). For all cases, the $f_{r_{ice}}$ is narrower than the corresponding f_{r_w} since the number density of cloud droplets exceeds that of ice crystals.

5 Conclusions

We have studied mixed-phase cloud glaciation using a three-dimensional DNS model with Lagrangian microphysics. We determine that flow turbulence influences the mean and fluctuating thermodynamic properties, microphysics of cloud droplets and ice crystals, and thus glaciation. Also, the number density, size, and geometry of ice crystals have impact on glaciation, depending on the details of simulations. The findings presented in this letter intend to improve our microscale understanding of the glaciation process. Better estimation of the ice and liquid masses in turbulent mixed-phase clouds will contribute to reducing uncertainty in the prediction of cloud albedo and thereby Earth's radiation budget.

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Open Research

The data that support the findings of this work are available from the corresponding author upon reasonable request.

References

- Bailey, M. & Hallett, J. (2004), Growth rates and habits of ice crystals between – 20 and – 70 °C. *Journal of the Atmospheric Sciences*, 61(5), 514-544. doi: 10.1175/1520-0469(2004)061<0514:GRAHOI>2.0.CO;2.
- Chen, J.P. & Lamb, D. (1994), The theoretical basis for the parameterization of ice crystal habits: Growth by vapor deposition. *Journal of Atmospheric Sciences*, 51(9), 1206-1222. doi: 10.1175/1520-0469(1994)051<1206:TTBFTP>2.0.CO;2.
- Chen, S., Xue, L., Tessendorf, S., Ikeda, K., Weeks, C., Rasmussen, R., Kunkel, M., Blestrud, D., Parkinson, S., Meadows, M. & Dawson, N. (2023), Mixed-phase direct numerical simulation: ice growth in cloud-top generating cells. *Atmospheric Chemistry and Physics*, 23(9), 5217-5231. doi: 10.5194/acp-23-5217-2023.
- Hill, A.A., Field, P.R., Furtado, K., Korolev, A. & Shipway, B.J. (2014), Mixed-phase clouds in a turbulent environment. Part 1: Large-eddy simulation experiments. *Quarterly Journal of the Royal Meteorological Society*, 140(680), 855-869. doi: 10.1002/qj.2177.
- Huang, J. (2018), A simple accurate formula for calculating saturation vapor pressure of water and ice. *Journal of Applied Meteorology and Climatology*, 57(6), 1265-1272. doi: 10.1175/JAMC-D-17-0334.1.
- Korolev, A. (2007), Limitations of the Wegener–Bergeron–Findeisen mechanism in the evolution of mixed-phase clouds. *Journal of the Atmospheric Sciences*, 64(9), 3372-3375. doi: 10.1175/JAS4035.1.
- Korolev, A.V., Isaac, G.A., Cober, S.G., Strapp, J.W. & Hallett, J. (2003), Microphysical characterization of mixed-phase clouds. *Quarterly Journal of the Royal Meteorological Society: A journal of the atmospheric sciences, applied meteorology and physical oceanography*, 129(587), 39-65. doi: 10.1256/qj.01.204.
- Morrison, H., De Boer, G., Feingold, G., Harrington, J., Shupe, M.D. & Sulia, K. (2012), Resilience of persistent Arctic mixed-phase clouds. *Nature Geoscience*, 5(1), 11-17. doi: 10.1038/ngeo1332.
- Pinsky, M., Khain, A. & Korolev, A. (2015), Phase transformations in an ascending adiabatic mixed-phase cloud volume. *Journal of Geophysical Research: Atmospheres*, 120(8), 3329-3353. doi: 10.1002/2015JD023094.
- Sarnitsky, G., Sardina, G., Svensson, G., Pumir, A., Hoffmann, F. & Mehlig, B. (2025), Does small-scale turbulence matter for ice growth in mixed-phase clouds?. *Physical Review Fluids*, 10(5), 053803. doi: 10.1103/PhysRevFluids.10.053803.
- Sharfuddin, A.A.M., Ladeinde, F., Liu, Y., Yang, F., Zhang, T., Atif, M., Lin, M. & Lopez-Marrero, V. (2025), Direct numerical simulations of activation and deactivation in turbulent atmospheric clouds. *Physics of Fluids*, Under review.
- Sheridan, L.M., Harrington, J.Y., Lamb, D. & Sulia, K. (2009), Influence of ice crystal aspect ratio on the evolution of ice size spectra during vapor depositional growth. *Journal of the atmospheric sciences*, 66(12), 3732-3743. doi: 10.1175/2009JAS3113.1.
- Storelvmo, T. & Tan, I. (2015), The Wegener-Bergeron-Findeisen process—Its discovery and vital importance for weather and climate. *Meteorologische Zeitschrift*, 24(4), 455-461. doi: 10.1127/metz/2015/0626.
- Sulia, K.J. & Harrington, J.Y. (2011), Ice aspect ratio influences on mixed-phase clouds: Impacts on phase partitioning in parcel models. *Journal of Geophysical Research: Atmospheres*, 116(D21). doi: 10.1029/2011JD016298.